

COM3502-4502-6502 SPEECH PROCESSING

Lecture 2 Sound

Sounds Everywhere



What is Sound?

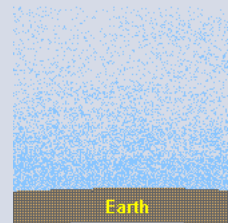
- Acoustic energy
- A mechanical wave
- Vibrations that travel through the air (*or another medium*)
- Variation in (air) **pressure**



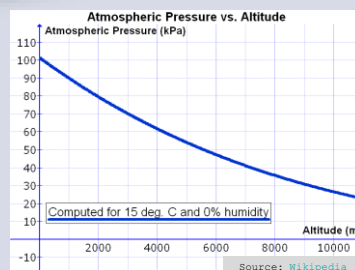
Source: Movie 'X-Men Origins: Wolverine' © Marvel

Air Pressure

Density of
air
molecules



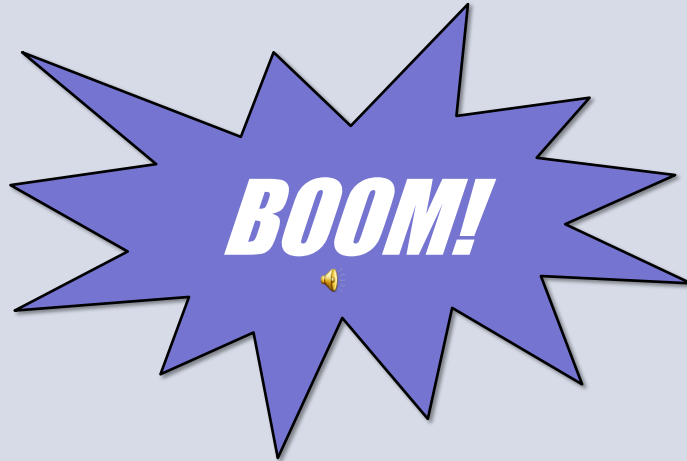
2 miles
↑
decreasing
air pressure
↑
sea level



Measured in:

- atmospheres
- pascals
- millibars
- mmHg

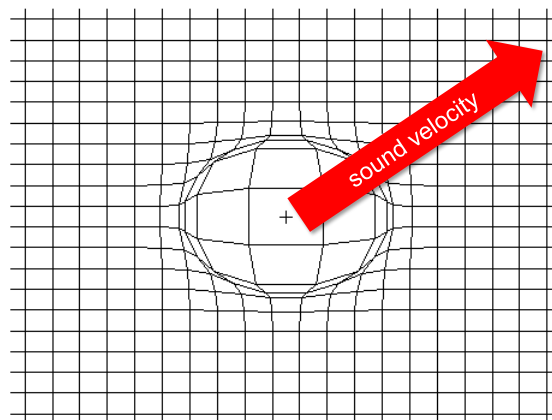
Sound Pressure



Sound Pressure

Speed of propagation depends on the density of the medium

- 343 m/s
(in air)
- 1372 m/s
(in water)
- 5182 m/s
(in steel)



Picture source: http://commons.wikimedia.org/wiki/File:Ondes_compression_2d_20.gif

Sound Pressure Level (SPL)

- Sound pressure is the **local pressure deviation** from the ambient atmospheric pressure caused by a sound wave
- The SI unit for sound pressure p is the *pascal* (Pa)
- Sound pressure level (SPL) is a logarithmic measure of the sound '**amplitude**' relative to a reference value

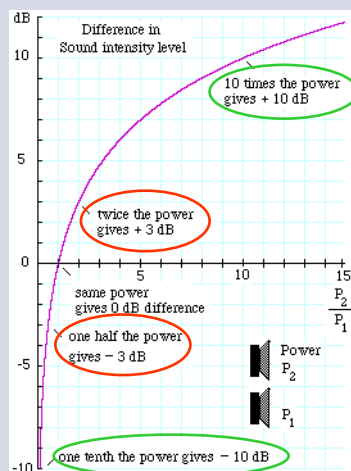
$$L_p = 20 \log_{10} \left(\frac{p_{rms}}{p_{ref}} \right) \text{ dB}$$

- 0 dB SPL has been defined to be the threshold of human hearing ($20 \mu\text{Pa}$)

What is dB?



dB	Power ratio
100	10,000,000,000
90	1,000,000,000
80	100,000,000
70	10,000,000
60	1,000,000
50	100,000
40	10,000
30	1,000
20	100
10	10
6	$3.981 \approx 4$
3	$1.995 \approx 2$
1	1.259
0	1
-1	0.794
-3	$0.501 \approx \frac{1}{2}$
-6	$0.251 \approx \frac{1}{4}$
-10	0.1
-20	0.01
-30	0.001



dB is the **ratio** of two values

$$10 \log_{10} \left(\frac{P_1}{P_2} \right)$$

- Doubling** power $\approx 3 \text{ dB}$
 $10 \log_{10}(2) \approx 3 \text{ dB}$
- 10 times** the power $\approx 10 \text{ dB}$
 $10 \log_{10}(10) \approx 10 \text{ dB}$

Picture: <https://www.animations.physics.unsw.edu.au/jw/dB.htm>

What is dB?



dB	Power ratio	Amplitude ratio
100	10,000,000,000	100000
90	1,000,000,000	31623
80	100,000,000	10000
70	10,000,000	3162
60	1,000,000	1000
50	100,000	316.2
40	10,000	100
30	1,000	31.62
20	100	10
10	10	3.162
6	3.981 ≈ 4	1.995 ≈ 2
3	1.995 ≈ 2	1.413 ≈ √2
1	1.259	1.122
0	1	1
-1	0.794	0.891
-3	0.501 ≈ ½	0.708 ≈ √½
-6	0.251 ≈ ¼	0.501 ≈ ½
-10	0.1	0.3162
-20	0.01	0.1
-30	0.001	0.03162

dB is the **ratio** of two values

- Power values $10 \log_{10} \left(\frac{P_1}{P_2} \right)$

$$L_P = 10 \log_{10} \left(\frac{p_{rms}^2}{p_{ref}^2} \right) \text{dB}$$

$$L_P = 10 \log_{10} \left(\left(\frac{p_{rms}}{p_{ref}} \right)^2 \right) \text{dB}$$

- Level / amplitude values

$$L_P = 20 \log_{10} \left(\frac{p_{rms}}{p_{ref}} \right) \text{dB}$$

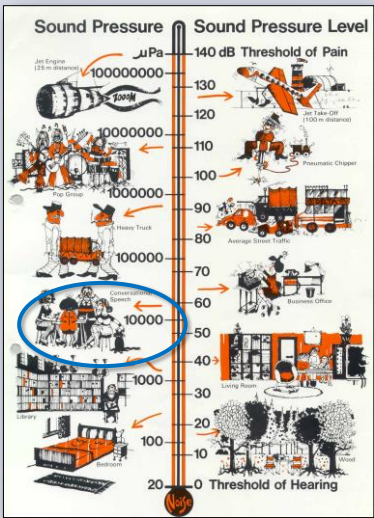
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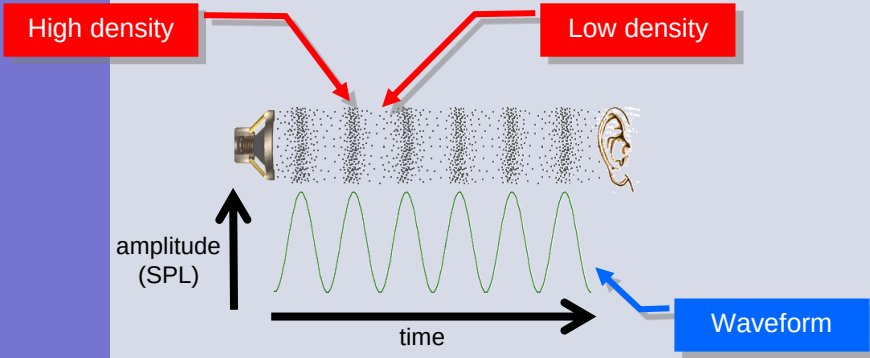
Sound Pressure Level (SPL)



SPL meter

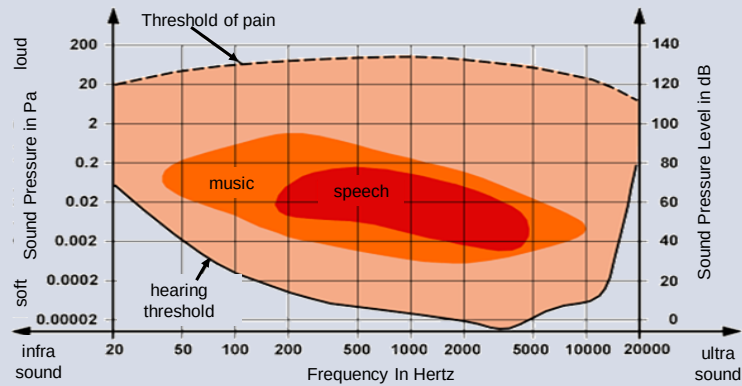
<http://personal.cityu.edu.hk/~bsapplec/sound.htm>

Sound Pressure Waveform



Picture source: <http://www.mediacollege.com/audio/01/sound-waves.html>

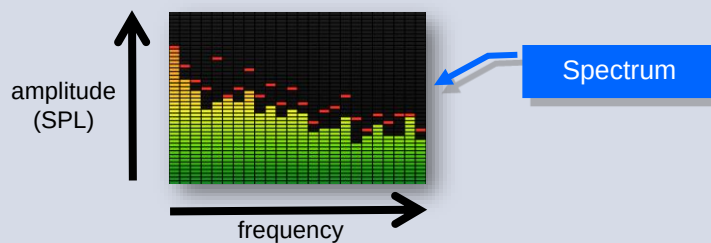
Sound Pressure over Frequency



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The Sound 'Spectrum'

- A sound may be perceived as having 'pitch' and 'timbre'
- Pitch relates to the frequency at which repetitions occur (*the number of occurrences of a repeating event per unit time*)
- Timbre relates to the SPL at different frequencies



Source: <http://appfinder.lissoft.com/app/led-audio-spectrum.html>

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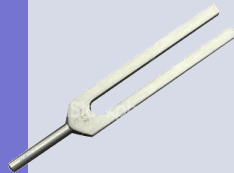
Sound Frequency

- Frequency is expressed as 'cycles per second' (Hertz)

$$f = \frac{n}{t} = \frac{1}{T} \text{ Hz}$$

... where **n** is the number of cycles in time **t**, and **T** is the 'period' (the time of one cycle in seconds)

For example, if 22 events occur in 50 msec ...

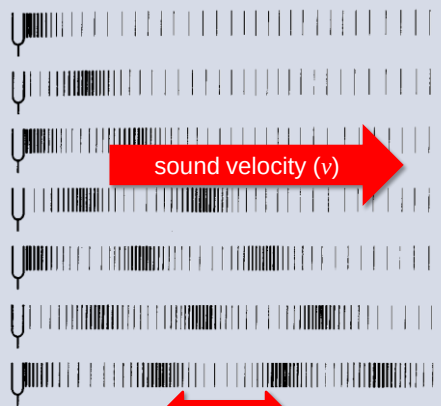


$$f = \frac{22}{0.05} = 440 \text{ Hz}$$

$$T = \frac{1}{440} = 2.27 \text{ msec}$$



Sound Frequency



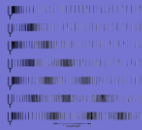
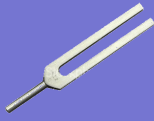
$$\lambda = \frac{v}{f} \text{ in m}$$

$$f = \frac{v}{\lambda} \text{ in Hz}$$

Explains the 'Doppler' effect



<http://personal.cityu.edu.hk/~bsapplec/natureof1.htm>



Sound Frequency

For example, for a tuning fork vibrating at 440 Hz in air, the '**wavelength**' of the resulting sound is ...

$$\lambda = \frac{v}{f} = \frac{343}{440} = 0.779 \text{ m}$$

... where the speed of sound in air v is 343 m/sec.

In water the speed of sound v is faster (1372 m/sec), so the wavelength λ is longer (3.117 m).

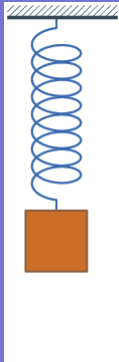


Types of Sound

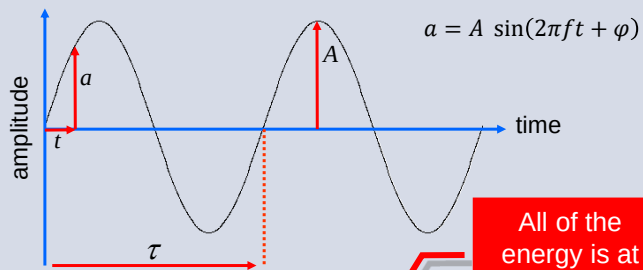
- '**Impulsive**'
(e.g. an explosion or a handclap)
- '**Noisy**'
(e.g. the wind or the surf on a beach)
- '**Repetitive**'
(e.g. a buzzing insect or a tuning fork)

These reflect the different ways in which sound pressure can vary over *time*

A Pure Tone

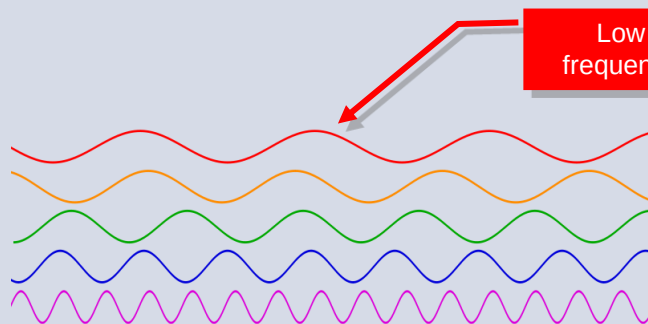


- Recall that frequency corresponds to the number of occurrences of a repeating event per unit time
- The simplest repeating signal is a '**sinusoid**' (as produced by a weight on a spring, a tuning fork or a whistle)



- A sine wave sound is known as a '**pure tone**' (with frequency f , amplitude A and phase ϕ)

All of the energy is at one frequency



High frequency

Pure tones at different frequencies

Superposition

- Sound waves travelling through a medium **'interfere'** with each other
- However, when sound waves combine, they obey the **'superposition principle'** (*i.e., they simply add*)
- If two pure tones have the same frequency, amplitude and phase, the result is a pure tone at the same frequency with twice the amplitude (**'constructive interference'**)
- If two pure tones have the same frequency and amplitude, but opposite phase, they cancel each other out (**'destructive interference'**)

Resonance

- When waves encounter an obstacle, they are reflected
- Direct and reflected waves combine (*using the principle of superposition*)
 - constructive interference results in **'resonance'** (*i.e. the emphasis of energy at particular frequencies*)
 - destructive interference results in **'anti-resonance'** (*i.e. the reduction of energy at particular frequencies*)
- Acoustic enclosures have resonant characteristics that are a function of their ...
 - physical dimensions
 - surface reflection properties
- A simple acoustic structure (*such as a tube*) has straightforward resonant characteristics



Resonant Tubes

- The resonant frequency f of a cylinder open at both ends is given by ...

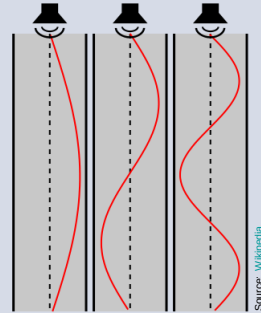
$$f = \frac{i v}{2 L}$$

where ...

v is the speed of sound

L is the length of the tube

i is a positive integer (1, 2, 3 ...)



Source: Wikipedia

- $i = 1$ gives the frequency of the 'fundamental'
- $i = 2, 3 \dots$ gives the frequencies of the 'harmonics'

Resonant Tubes

- The resonant frequency f of a cylinder open at one end is given by ...

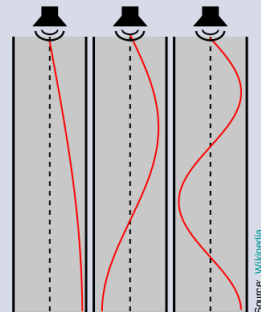
$$f = \frac{(2 i - 1) v}{4 L}$$

where ...

v is the speed of sound

L is the length of the tube

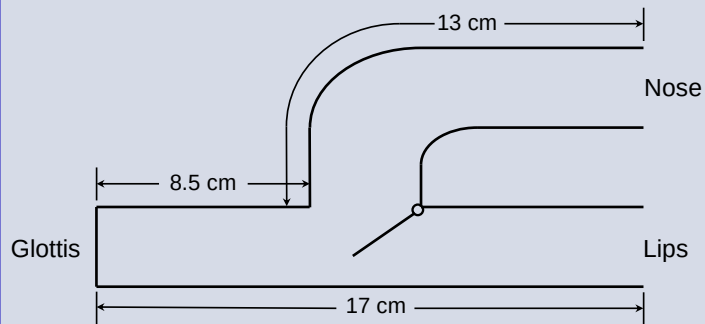
i is a positive integer (1, 2, 3 ...)



Source: Wikipedia

- $i = 1$ gives the frequency of the 'fundamental'
- $i = 2, 3 \dots$ gives the frequencies of the 'harmonics'

Simplified Model of the Vocal Tract

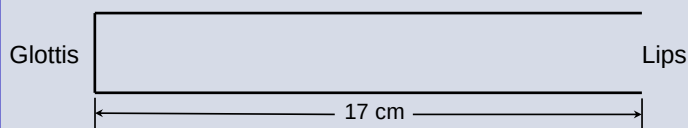


Adapted from: [Vary, Martin 2006]

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Simplified Model of the Vocal Tract

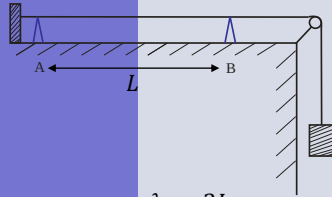
- We consider the nose cavity to be closed by the velum tube is closed at one end (the glottis end) and open at the other end (the mouth end)
 - ▶ Resonances of a tube



Adapted from: [Vary, Martin 2006]

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Resonances of a string and a tube



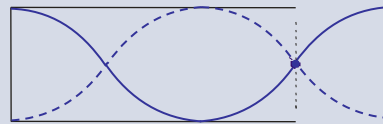
$$\lambda_1 = 2L$$



$$\lambda_3 = \frac{2L}{3}$$

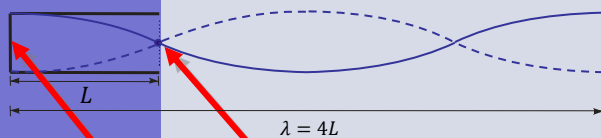


- wavelength creating 1st resonance: $\lambda_1 = 2L$
- In general:
 - wavelength is $\lambda_i = \frac{2L}{i}$ for harmonic i or $f_i = \frac{iv}{2L}$
- Side conditions (fixations at points A and B) define where minima and maxima are for standing waves



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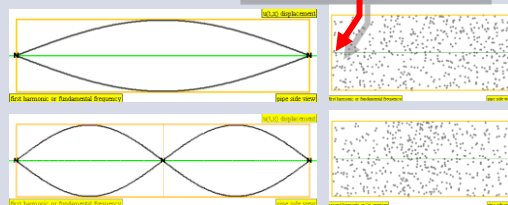
Harmonics of a tube



closed
Pressure max.

open

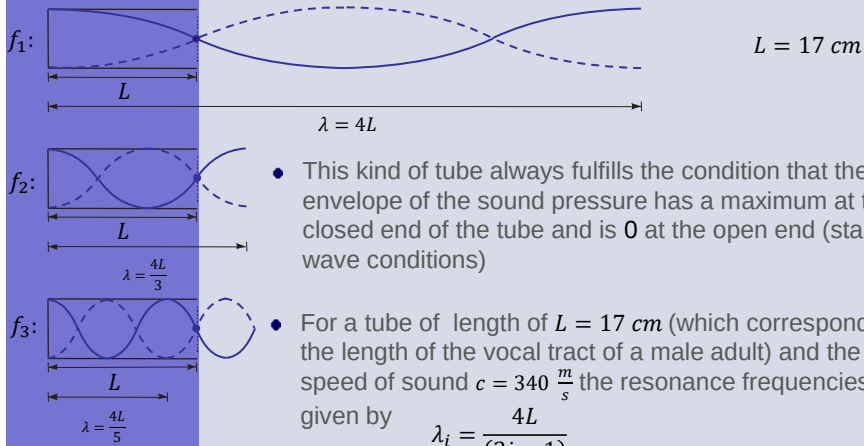
closed
displacement min.



For a tube of length of $L = 17 \text{ cm}$ (which corresponds to the length of the vocal tract of a male adult)

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Harmonics of a tube



$$\lambda_i = \frac{4L}{(2i-1)}$$

$$f_i = \frac{(2i-1)c}{4L} = (2i-1) 500 \text{ Hz} = \{500, 1500, 2500, 3500, \dots\} \text{ Hz}$$

$$i = 1, 2, 3, \dots$$

Resonant Tubes

- In the general case, the resonant frequency f of a cylinder is given by ...

$$f = \frac{(2i - (x_1 + x_2)) c}{4L}$$

where ...

c is the speed of sound
(symbol c from the Latin *celeritas* meaning "velocity")

L is the length of the tube

i is a positive integer (1, 2, 3 ...)

x_1 is the degree of opening at one end of the tube

x_2 is the degree of opening at the other end of the tube

- $i = 1$ gives the frequency of the 'fundamental'
- $i = 2, 3 \dots$ gives the frequencies of the 'harmonics'

Resonant Tubes

E.g. for a **25 cm** cylinder ...

- open at both ends
 - fundamental = 686 Hz
 - 1st harmonic = 1372 Hz
 - 2nd harmonic = 2058 Hz
 - 3rd harmonic = 2744 Hz
 - etc.
- open at one end, closed at the other
 - fundamental = 343 Hz
 - 1st harmonic = 1029 Hz
 - 2nd harmonic = 1715 Hz
 - 3rd harmonic = 2401 Hz
 - etc.
- closed at both ends
 - fundamental = 0 Hz
 - 1st harmonic = 686 Hz
 - 2nd harmonic = 1372 Hz
 - 3rd harmonic = 2058 Hz
 - etc.

$$f = \frac{(2i - (x_1 + x_2))c}{4L}$$

$$f = \frac{(2i - (0 + 0)) 343 \frac{m}{s}}{1 m}$$

$$f = \frac{(2i - (0 + 1)) 343 \frac{m}{s}}{1 m}$$

$$f = \frac{(2i - (1 + 1)) 343 \frac{m}{s}}{1 m}$$

Resonant Tubes

E.g. for a **12.5 cm** cylinder ...

- open at both ends
 - fundamental = 1372 Hz
 - 1st harmonic = 2744 Hz
 - 2nd harmonic = 4116 Hz
 - 3rd harmonic = 5488 Hz
 - etc.
- open at one end, closed at the other
 - fundamental = 686 Hz
 - 1st harmonic = 2058 Hz
 - 2nd harmonic = 3430 Hz
 - 3rd harmonic = 4802 Hz
 - etc.
- closed at both ends
 - fundamental = 0 Hz
 - 1st harmonic = 1372 Hz
 - 2nd harmonic = 2744 Hz
 - 3rd harmonic = 4166 Hz
 - etc.

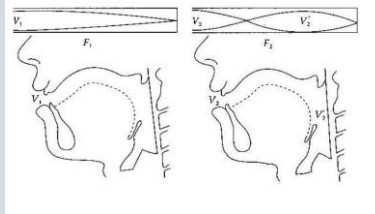
What happens if we shorten the cylinder to half its length?

$$f = \frac{(2i - (x_1 + x_2))c}{4L}$$

All the resonant frequencies are double those for a 25 cm tube.

The Human Vocal Tract

- In the **neutral position** ("uh"), the adult male vocal tract is equivalent to a **17.5 cm tube** open at one end (*the mouth*) and closed at the other (*the larynx*)

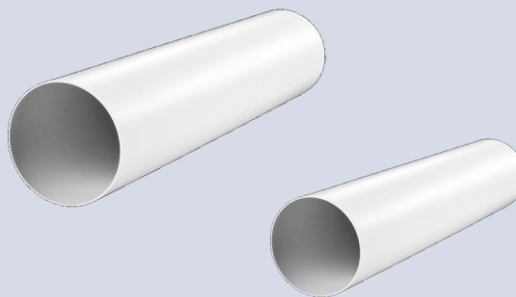


$$f = \frac{(2i - (x_1 + x_2))c}{4L}$$

- This means that it has the following **resonant** characteristics ...
 - fundamental = 490 Hz
 - 1st harmonic = 1470 Hz
 - 2nd harmonic = 2450 Hz
 - 3rd harmonic = 3430 Hz
 - etc.



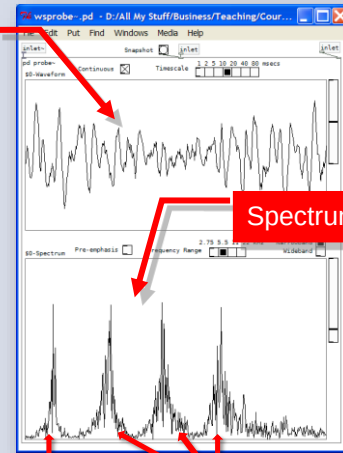
Resonant Tubes



plastic
wastepipe

Resonant Tubes

Waveform



Spectrum

fundamental

harmonics

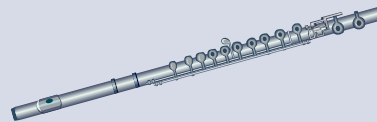
Excitation

- In order for an acoustic structure to resonate it needs to be '**excited**' (driven by a '**source**' of energy)
- As we've already seen, sound sources can be ...
 - impulsive
 - noisy
 - repetitive
- These are all rich in energy at many different frequencies
- So the effect of resonances/anti-resonances is to enhance/reduce the incoming energy at different frequencies
- This can be thought of as '**filtering**'

Excitation

The three modes of excitation
(*impulsive, noisy, repetitive*)
give rise to three classic types
of musical instruments...

- percussion (e.g., a drum)
- wind (e.g., a flute)
- string (e.g., a violin)



Excitation

What type of
excitation
source is used
for each of
these musical
instruments
(*impulsive,
repetitive,
noisy*)?

impulsive



noisy

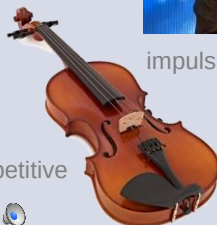


repetitive



impulsive+repetitive+noisy

repetitive



impulsive



This lecture has covered ...

- Air pressure
- Sound pressure
- Sound pressure level (SPL)
- Impulsive/noisy/repetitive sounds
- Sound frequency
- Sound waves
- Resonance & resonant tubes
- The human vocal tract
- Excitation

Any Questions ?

*Raise during lecture, or post in the Blackboard
Discussion group*



Next time ...

Signals